

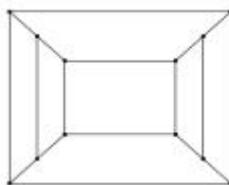
RMO– 2014(4)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABCD be an isosceles trapezium having an incircle; let AB and CD be the parallel sides and let CE be the perpendicular from C on to AB. Prove that CE is equal to the geometric mean of AB and CD.
 2. If x and y are positive real numbers, prove that $4x^4 + 4y^3 + 5x^2 + y + 1 \geq 12xy$.
 3. Determine all pairs $m > n$ of positive integers such that :
 $1 = \gcd(n+1, m+1) = \gcd(n+2, m+2) = \dots = \gcd(m, 2m-n)$.
 4. What is the minimal area of a right angled triangle whose inradius is 1 unit?
 5. Let ABC be an acute angled triangle and let I be the incentre. Let the incircle of triangle ABC touch BC in D. The incircle of the triangle ABD touches AB in E; the incircle of the triangle ACD touches BC in F. Prove that B, E, I, F are concyclic.
 6. In the given figure, can the numbers 1, 2, 3, 4, ..., 18 be placed one on each line segment, such that the sum of the numbers on the three line segments meeting at each point is divisible by 3?



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